# Johannes Roth

PhD Candidate in Computational Cognitive Neuroscience

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## Summary

PhD candidate at MPI CBS and University of Gießen specializing in computational cognitive neuroscience. Research focuses on improving fMRI data acquisition efficiency for large-scale visual phenotyping studies by investigating high-frequency recording and optimal stimulus selection strategies. Experienced in designing/analyzing fMRI experiments, applying machine learning and computational modeling to study the human visual system.

## Education

**May 2022 – Present. PhD Candidate**, Max Planck Institute for Human Cognitive and Brain Sciences (Leipzig) & University of Gießen
Investigating methods to enhance fMRI data acquisition efficiency for large-scale deep-phenotyping studies. Research focuses on high-frequency stimulus presentation and optimal stimulus pre-selection strategies. Designing and analyzing fMRI experiments, developing computational models, and applying machine learning techniques to study the human visual system. Supervised by Prof. Dr. Martin Hebart.

**2017 – 2021. M.Sc. Computer Science**, Leipzig University
Grade: 1.2 (Sehr Gut)
Focus areas: Data Science, Machine Learning, Medical Image Processing, Neuroscience.
Thesis: *Synthesizing preferred stimuli for voxels in the visual system.* Supervised by Prof. Dr. M. Hebart & Dr. K. Seeliger.

**2014 – 2017. B.Sc. Business Information Systems**, Leipzig University
Grade: 1.5 (Sehr Gut)
Thesis: *Visualisierung der Struktur von ABAP-Software mittels generativer Softwarevisualisierung.* Supervised by Prof. Dr. U. Eisenecker & Dr. Pascal Kovacs.

## Publications

**2025** M. Badwal, J. Bergmann, **J. Roth**, C. Doeller, M. Hebart. *The scope and limits of fine-grained image and category information in the ventral visual pathway*, Journal of Neuroscience, 45(3).

**2023 J. Roth**, Y. Miyawaki, M. Hebart. *Assessing high stimulus presentation rates for fMRI studies*, Journal of Vision (VSS Abstract), 23(9), 5070–5070.

**2021 J. Roth**, J. Keller, S. Franke, T. Neumuth, D. Schneider. *Multi-plane UNet++ ensemble for glioblastoma segmentation*, MICCAI BrainLesion Workshop (BrainLes), LNCS 12929, pp. 285–294.

**2021 J. Roth**, K. Seeliger, T. Schmid, M. Hebart. *Synthesizing preferred stimuli for voxels in the visual system*, Poster at Computational and Systems Neuroscience (Cosyne) 2021.

## Work Experience

**Jun 2021 – May 2022. Research Assistant (ML in Medicine)**, ScaDS.AI Dresden/Leipzig
Developed and improved deep learning models for medical applications. Focused on enhancing uncertainty-aware prostate cancer mortality prediction, achieving state-of-the-art results, and building models for automated brain tumor segmentation.

**Oct 2020 – May 2021. Full-stack Developer (Freelance)**, Kimetric UG
Developed two websites for academic clients. Configured hosting (Linux, Nginx, Gunicorn), built Django backends for site/user management, and created frontends using HTML, CSS, and JavaScript.

**Oct 2019 – May 2021. Data Scientist (Working Student)**, CHECK24 Vergleichsportal
Led the development of an image processing micro-service, including data acquisition, training ML models (deduplication, retrieval, classification, upscaling, quality assessment), and building a Python (Flask/Gunicorn) backend. Improved search result relevance by applying Bayesian optimization to recommender systems and developing ranking models. Conducted statistical analyses (e.g., outlier detection, time series visualization) and prepared results for stakeholders. Briefly contributed to backend API development (PHP, Go).

**Oct 2018 – Oct 2019. Data Scientist (Working Student)**, Webdata Solutions GmbH (now part of Vistex)
Led the development of a product image retrieval and deduplication pipeline. Tasks included crawling and cleaning large image datasets, and training models for image representation, segmentation, and text classification using TensorFlow and PyTorch. This system significantly outperformed previous methods and formed the basis for a research grant.

**Feb 2018 – Aug 2018. Data Analyst (Working Student)**, Mercateo Services GmbH (now Unite)
Assisted in developing cloud-based data warehousing solutions (AWS, Apache NiFi) and ETL processes. Contributed to building integration test infrastructure (Python) and reporting systems (Pentaho, Java, R).

**Apr 2017 – Sep 2018. Academic Tutor**, Universität Leipzig
Tutored the undergraduate course 'Developing Distributed Systems' (covering IPC, Java EE, REST APIs). Presented course material, led exercise sessions, corrected assignments, and answered student questions for two semesters.

**Oct 2016 – Mar 2017. Intern (SAP Development)**, GISA GmbH
Interned in the SAP development (metering) team. Developed ABAP message transports and created a SAP Fiori application prototype (MVP) for electric metering field services.

## Volunteer Work

**2023 – 2024. Internal PhD Representative**, MPI CBS
Represented PhD student interests in institutional policy discussions and administrative matters.

## Skills

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| **Programming:** | Python, SQL, Git, Bash, MATLAB, Java, R, JavaScript, HTML/CSS |
| **ML/Data Science:** | Scikit-learn, Pandas, NumPy, SciPy, TensorFlow, PyTorch, Keras, Statsmodels, Machine Learning (Classification, Regression, Clustering, Dimensionality Reduction), Deep Learning (CNNs, GANs, Transformers), Computer Vision, Statistical Analysis, Data Visualization, Bayesian Optimization |
| **Neuroimaging:** | fMRI Experiment Design & Analysis (SPM, FSL, Nilearn, BrainStat), Neural Decoding, Encoding Models, Representational Similarity Analysis (RSA), Computational Modeling |
| **Tools/Platforms:** | Docker, Linux, High-Performance Computing (SLURM), AWS (basic), Nginx, Gunicorn, Flask, Django, Apache NiFi, SAP (ABAP, Fiori) |
| **Languages:** | German (Native), English (Fluent) |